

Piyush Hinduja

Salt Lake City, UT | (385) 507-6190 | piyushinduja20@gmail.com

Portfolio: <https://piyushinduja.github.io/Portfolio/> | LinkedIn: [linkedin.com/in/piyush-hinduja/](https://www.linkedin.com/in/piyush-hinduja/)

EDUCATION

University of Utah, Salt Lake City, Utah
Master of Science (MS) Computer Science

August 2023 - May 2025
GPA: 3.9/4.0

University of Mumbai, Bandra, India
Bachelor of Engineering (BE) Computer Engineering

July 2019 - May 2023
GPA: 3.84/4.0

SKILLS

Languages: *Expert (5+ yrs):* Python, Java, JavaScript *Intermediate (3+ yrs):* C, C++ *Beginner (1+ yrs):* Kotlin, SQL, Shell

Machine Learning and Data Processing: PyTorch, Transformers, TensorFlow, Langchain, Scikit-Learn, Pandas, PySpark, OpenCV

Cloud & Databases: AWS, Amazon RDS, CloudLab, Docker, Kubernetes, MySQL, MongoDB, ChromaDB

Web Development: Node.js, React.js, TypeScript, HTML, XML, CSS, RESTful APIs

Miscellaneous: Data Structures, CI/CD, Git, JIRA, Agile, SCRUM

WORK EXPERIENCE

Data Analyst | University of Utah, Salt Lake City, Utah

March 2024 - Present

- Architected scalable **data processing pipeline** using Python's multiprocessing and concurrent.futures library to parallelize file processing, API requests, and text cleaning operations, **reducing per-filing processing time by 40%**
- Implemented advanced **NLP analysis using transformer models (DeBERTa, FinBERT, RoBERTa, etc.)** and **PyTorch** for financial document processing, improving forward-looking statement identification by 27%
- Developed robust **prompt engineering solutions** to integrate multiple AI models (OpenAI, Anthropic, Gemini) for automated financial text analysis, **comparing AI-generated insights against traditional algorithms**
- Optimized financial data processing workflows, achieving **faster execution and lower memory consumption** through optimized data structures and parallelized computations

Machine Learning Researcher | FLUX Research Group, Salt Lake City, Utah

August 2024 - Present

- Engineered automated deployment system using **Python to streamline NVIDIA/CUDA toolkit** integration on high-performance computing clusters, **cutting setup time by 90%** for 50+ researchers
- Collaborating with the Wireless Powder Testbed team to implement **predictive modeling solutions** focusing on enhancing experimental outcomes through machine learning applications for a **network of 10000+ UEs** (User Equipments)

Full Stack Developer Intern | Exposys Data Labs, Bangalore, India

June 2021 - July 2021

- Spearheaded the launch of a **tourism project**, integrating key features to **enhance user experience** and deliver valuable information about 50+ Indian travel destinations
- Utilized **MERN stack** for an efficient frontend and backend integration, creating a **responsive, user-friendly interface** and enabling content management

PROJECTS

Road Damage Detection | (~120 hours) | Object-Detection, YOLOv5, FasterRCNN

[GitHub Link](#)

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5** and **FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the **RDD-2020 dataset**
- Achieved model accuracies of **52.67%** with YOLOv5 and **44.45%** with FasterRCNN by training and **fine-tuning** on different **hyper-parameters**
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping **lower the frequency of accidents and fatalities**

Dependency Parser | (~60 hours) | PyTorch, GloVe Embeddings

[GitHub Link](#)

- Streamlined a Transition based Dependency parser for the **university's NLP group**, inspired by "A Fast and Accurate Dependency Parser using Neural Networks" (**Chen and Manning, 2014**), returning a parse tree for a sentence in less than **100 ms**
- Focused on Data Transformation, Model Building and Fine-tuning over different embeddings and hyperparameters, achieving a **UAS (Unlabeled Attachment Score) of 0.76** and **LAS (Labeled Attachment Score) of 0.705**

Pizzeria | (~60 hours) | Web Development, MERN, Firebase

[GitHub Link](#)

- Designed a **pizza ordering web application** featuring live order updates, personalized carts, favorite pizza options, and an integrated chat feature for **real-time customer-store communication**
- Utilized a tech stack including **HTML5, CSS, JavaScript, Node.js, Express.js, MongoDB**, and libraries such as **Socket.io** for real-time updates and **Passport** for secure authentication

Language Modeling | (~40 hours) | NLP, RNN, N-gram, PyTorch

[GitHub Link](#)

- Developed **character-level LSTM model** in PyTorch to predict character sequences based on **50-character context windows**
- Implemented **teacher forcing** during training to prevent error propagation, using ground truth characters instead of predictions for subsequent timesteps
- Achieved **perplexity of 9.43** through systematic model architecture optimization and training methodology